

Everything Summer '26

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1 Kitaev Quantum Double Model

The flow: Fix a lattice $\rightarrow$ edges with quantum degrees of freedom $\rightarrow$ local operators and Hamiltonian $\rightarrow$ ground state and excitations.

State space: Lattice with V , E , and F : the set of vertices, edges, and faces, respectively. Assign an orientation to each $e \in E$ (arbitrary).

$$\text{Sites } \mathcal{S} = \{(p, v) \mid p \in F, v \in \partial p\}.$$

At every edge $e \in E$, we put a $|G|$ -dimensional qudit $\mathcal{H}_e = \mathbb{C}[G]$ with computational basis $\{|g\rangle\}$. Total Hilbert space $\mathcal{H}_{\text{tot}} = \otimes_{e \in E} \mathcal{H}_e$ with $\dim(\mathcal{H}_{\text{tot}}) = |G|^{|E|}$. Also,

$$|g_{e_1} g_{e_2} \dots g_{e_{|E|}}\rangle := |g_{e_1}\rangle \otimes |g_{e_1}\rangle \otimes \dots \otimes |g_{e_{|E|}}\rangle.$$

Note: The ring multiplication structure of $\mathbb{C}[G]$ comes into play in the local operators.

Hamiltonian: $A_g(v)$ is defined as follows: (Figure to be included)

$$|\dots g_1 g_2 g_3 g_4 \dots\rangle \mapsto |\dots gg_1 gg_2 g_3 g^{-1} g_4 g^{-1} \dots\rangle.$$

(Note that $A_{g_1} A_{g_2} = A_{g_1 g_2}$ since $g_2^{-1} g_1^{-1} = (g_1 g_2)^{-1}$.) We define the vertex operator $A(s)$ where $s \in \mathcal{S}$:

$$A(s) := A(v) = \frac{1}{|G|} \sum_g A_g(v)$$

This is a projector. *Proof:* $A^2(s) = \frac{1}{|G|^2} \sum_{g_1, g_2} A_{g_1} A_{g_2}(v) = \frac{1}{|G|^2} \sum_{g_1, g_2} A_{g_1 g_2}(v)$.

Now, due to the rearrangement of finite groups, $\frac{1}{|G|^2} \sum_{g_1, g_2} A_{g_1 g_2}(v) = \frac{1}{|G|^2} \sum_{g_1} \left(\sum_g A_g(v) \right) = \frac{1}{|G|^2} \cdot |G| \cdot \sum_g A_g(v) = A(v)$. $\square$

$B_h(v)$ is defined as follows: (Figure to be included)

If $h = e$, the relation $g^{o(e_1)} g^{o(e_2)} g^{o(e_3)} g^{o(e_4)} = e$ can be cyclically permuted ($o(e)$ is the orientation of edge e). Hence, the starting vertex for traversing the plaquette p **does not** matter.

$A_g(s)$ and $B_h(s)$ do not commute but satisfy

$$A_g(s) B_h(s) = B_{ghg^{-1}}(s) A_g(s). \quad (1)$$

In a more convenient notation, $A_g(v) B_h(p) = B_{ghg^{-1}}(p) A_g(v)$. This means that we start traversing the plaquette p counterclockwise from vertex v . (Figure to be included)

Firstly,

$$\begin{aligned} A_g(v) B_h(p) |\dots g_1 g_2 g_3 g_4 \dots\rangle &= \delta_{h, g_2^{-1} g_3 g_4 g_1^{-1}} A_g(v) |\dots g_1 g_2 g_3 g_4 \dots\rangle \\ &= \delta_{h, g_2^{-1} g_3 g_4 g_1^{-1}} A_g(v) |\dots gg_1 gg_2 g_3 g^{-1} g_4 g^{-1} \dots\rangle. \end{aligned}$$

Here, we ignore the effects of $A_g(v)$ on the rest of the vertices since they are irrelevant in this context. Secondly,

$$\begin{aligned}
& B_{ghg^{-1}}(p)A_g(v)|\dots g_1 g_2 g_3 g_4 \dots\rangle \\
&= B_{ghg^{-1}}(p)|\dots gg_1 gg_2 g_3 g^{-1} g_4 g^{-1} \dots\rangle \\
&= \delta_{ghg^{-1}, (g_2 g^{-1})^{-1} g_3 g_4 (gg_1)^{-1}} A_g(v)|\dots gg_1 gg_2 g_3 g^{-1} g_4 g^{-1} \dots\rangle.
\end{aligned}$$

Finally, we note that the conditions

$$\begin{aligned}
ghg^{-1} &= (g_2 g^{-1})^{-1} g_3 g_4 (gg_1)^{-1} \\
\Leftrightarrow ghg^{-1} &= g (g_2^{-1} g_3 g_4 g_1^{-1}) g^{-1} \\
\Leftrightarrow h &= g_2^{-1} g_3 g_4 g_1^{-1}
\end{aligned}$$

are all equivalent, meaning that

$$A_g(v)B_h(p) = B_{ghg^{-1}}(p)A_g(v) \quad \square$$

If we flip the direction of e_1 , we get $h = g_2^{-1} g_3 g_4 g_1$ and $ghg^{-1} = (g_2 g^{-1})^{-1} g_3 g_4 (g_1 g^{-1})$, which are also equivalent. The argument is similar if we flip e_2 .

They satisfy the same relations as the generators of DG, which implies that the algebra of local operators at a (?) site $s = (p, v)$ generated by A_g and B_h form an $N = |G|^{|E|}$ (the exponent $|E|$, for ‘a’ site, how?) dimensional matrix representation of the quantum double algebra DG given by the homomorphism

$$\begin{aligned}
\rho_s : DG &\rightarrow GL(N, \mathbb{C}) \\
D_{e, g} &\mapsto A_g(v) \\
D_{h, e} &\mapsto B_h(p).
\end{aligned}$$

Altogether, the quantum double **Hamiltonian** is given by

$$H = - \sum_{v \in V} A(v) - \sum_{p \in P} B(p), \quad (2)$$

where $B(p) := B_e(p)$. We can check from the non-commuting relation of A_g and B_h that all the projectors in the Hamiltonian commute with each other.